# Manual for uploading excel sheets to the centralized CGG meta data database

Version: 3, date: 2024-06-04

Author: Magnus Johannsen (magnus.johannsen@sund.ku.dk)

The intention for this manual is to guide the user in uploading as well as to introduce them to the processing, parsing, and formatting of their meta data.

## Contents

## Step-by-step guide to uploading

1. Log in and connect to the KU VPN

2. Go to http://dandyweb01fl.unicph.domain:5100 in you web browser of choice (Google Chrome is probably best, but others should work fine). This is the site you will use to upload your tabular/spreadsheet meta data to the tables in the database.

   The spreadsheets you upload must match the tables in the database. This means column names, data types and to some degree data format of your spreadsheets should be equivalent to that of the database table. If you encounter any issues, I've made a list of database compatible Excel sheets, that you will see on the website (the list of blue hyperlinks). If you use these templates, everything should work. Download the one that is relevant for you, by clicking on it. If you don't know which ones are, or if you encounter issues using them, contact the database administrator (see contact info below).

   Some of the sheets you can download may look like this:

   | Column Name | Data Type |
   |---|---|
   | Customer Name | Text |
   | Order Date | timestamp without time zone |
   | Order ID | Text |

   This means your spreadsheet should look like this:

   | Customer Name | Order Date | Order ID |
   |---|---|---|
   | text | timestamp without time zone | text |
   | text | timestamp without time zone | text |
   | text | timestamp without time zone | text |

3. Locate the sheet you want to upload and open it.

4. Save the sheet as "Unicode Text (.txt)" (windows) or "UTF-16 unicode text" (mac). Note: This will only save the active sheet, and not the whole Excel file, so make sure you are saving the correct sheet that you want to upload.

5. Go back to http://dandyweb01fl.unicph.domain:5100, click the "Choose File" button and choose the .txt file you just saved.

6. Select the spreadsheet type. This will ensure that your data is uploaded to the correct table, and that the database knows which sheet to expect (as described earlier, it is programmed to accept a fixed number of standard sheets).

7. Select the formatting options that are applicable to your sheet.

8. Press upload.

9. Your data will be parsed, meaning, dates and decimal points will be converted to the database standard (YYYY-MM-DD and ".") (see details below). If the formatting of your data

corresponds to the format options you chose, the parsed data will be displayed as a table. Review the data to make sure it is correct, and press confirm if it is.

10. When you press confirm, your data will be uploaded to the database and redirected to a page that shows you exactly how the data looks in the database. Double check that the data looks okay.

## Standard Excel format requirements/recommendations

Right now, the database is flexible in terms of format, but in the future, it will most likely be more rigid. I indicate if the format is currently required or recommended and I strongly recommend you comply with the recommended formatting, to prevent any tedious data cleanup in the future.

### Empty cells (highly recommended)

Leave cells empty if there is no data to put there. Do not write "N/A" or "-" or anything. Just leave it empty.

### Intervals (required)

Any reported intervals should be in a restricted mathematical notation and only the following characters are allowed: “[“ “)” “,” and numbers (no decimal allowed, unless you use “.” as decimal point). This means that [1, 2) is allowed but not (1, 2), (1, 2] or [1, 2]. This is unfortunately a standard the database follows that is difficult to change. If this is very difficult for you to comply with, contact the admin and we might be able to figure out something. However, the preferred method of reporting intervals is to have two columns (a “from” column and a “to” column).

### Decimal points (highly recommended)

The following is the safest and best approach for the database, as this is the standard of the database and therefore does not require error prone parsing:

Decimal point = "."

Thousand separators = "," or no separator (preferably the latter).

You can still upload other formats but try to change to this format pointing forward, as it has better compatibility with the database.

### Primary ids (requirement)

Primary ID columns cannot be empty. Remove rows that do not have a primary ID before uploading, or you will get an error. Primary IDs are defined as an identifier that uniquely identifies a row of the spreadsheet (archive sample ID, robot sample ID, library ID, field sample ID etc.).

### Multiple values in one cell (highly recommended)

If a column contains multiple values, separate them with semicolon. For example, if a sample was taken by multiple people the 'Sampled By' column input should be "Alice; Bob".

### Personal identifiers (highly recommended)

Full names. If you want to use initials, you need to upload a translator spreadsheet in this format:

| Initials | Full Name |
|---|---|
| MJ | Magnus Johannsen |
| MJK | Michael Jackson |
| MJD | Michael Jordan |

| | |
|---|---|
| MJG | Mick Jagger |

Select “Initials Translator” as spreadsheet type.

### Date (highly recommended)

Use ISO 8601 standard: YYYY-MM-DD (https://en.wikipedia.org/wiki/ISO_8601)

## Details on formatting and parsing:

When choosing a format from the dropdowns, any column will be validated against this format and the data will only be uploaded if it conforms to this format.

All data will first be parsed as text strings (a string of Unicode[1] characters) and if the format looks ok, it will be transformed to dates, floating point numbers, integers, ranges etc.

IMPORTANT: Be careful when choosing decimal point and thousand separator formats. If you choose wrong, there is no way for the parser to know if it is an error, and it will parse your data incorrectly. For example, if you have a small measurement of 1,9 and choose “.” as thousands separator the data will be interpreted as 1900 and not 1 point 9.

## Errors

During upload, you will most likely encounter errors. Most of the time, these are due to unexpected formatting, or data and should be easily fixed by yourself. However, if you encounter any errors that you cannot fix yourself, make sure to contact admin with the time and date of when the error happened and the traceback shown on the error page.

If you encounter a “Internal Server Error”, this means you encountered an error I did not anticipate. Please notify database administrator with the time and date for when it occurred, so I can investigate the error log.

## Admin contact info

Contact magnus.johannsen@sund.ku.dk if you encounter any errors you cannot resolve, or if you have any questions.

[1] Unicode characters are a standardized encoding system representing textual characters from multiple writing systems, enabling consistent representation of text across various platforms and languages.